



$$\left\{ \begin{array}{l} x^2 + y^2 = 2x + 2y \end{array} \right.$$

$$\left\{ \begin{array}{l} x^2 + y^2 = 2(1+a)x + 2(1-a)y - 2a^2 \end{array} \right.$$

$$\left\{ \begin{array}{l} (x-1)^2 + (y-1)^2 = 2 \\ (x-(1+a))^2 + (y-(1-a))^2 = 2 \end{array} \right.$$

$$(1+a; 1-a)$$

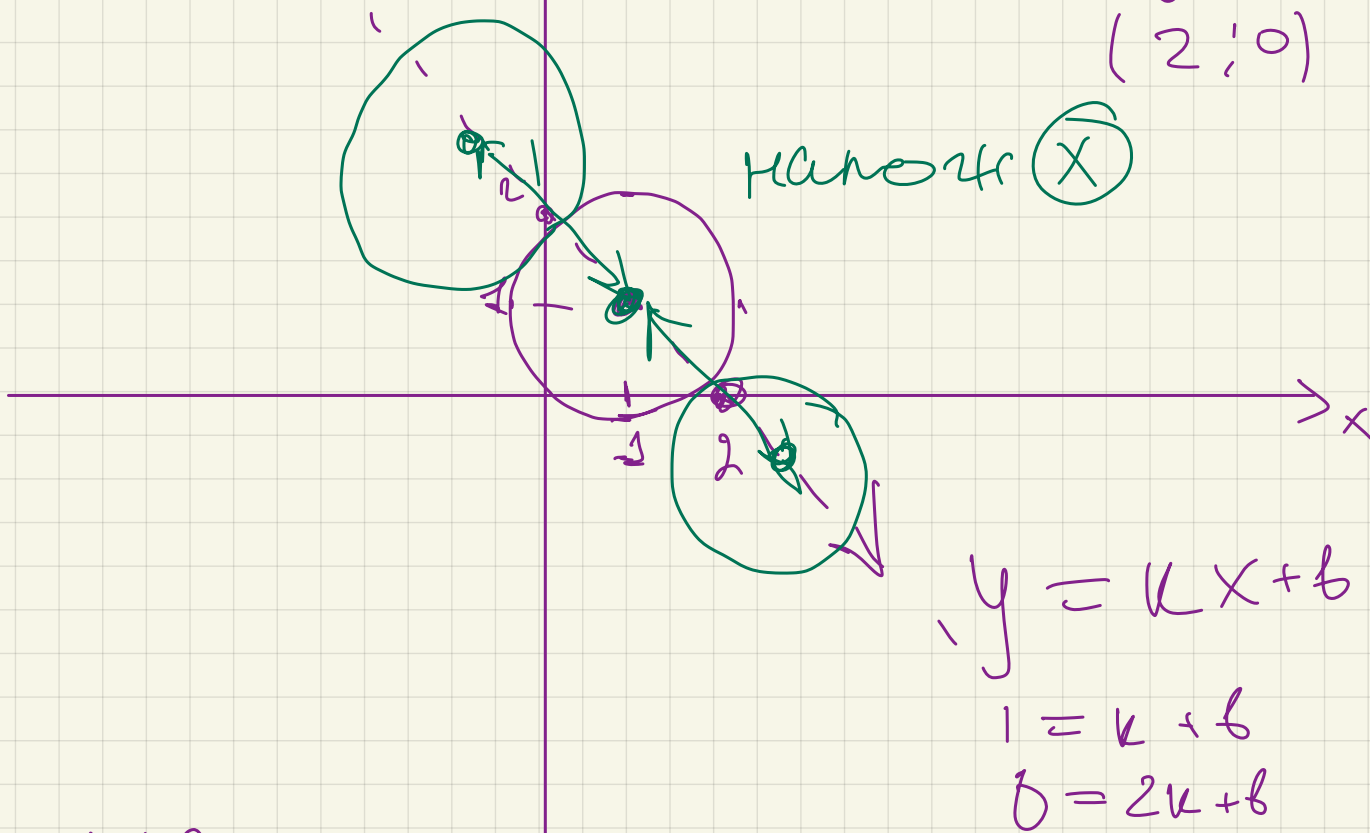
$$a=0$$

$$(1; 1)$$

$$a=1$$

$$(2; 0)$$

направление $\vec{x}$



$$y = -x + 2$$

$$y = kx + b$$

$$1 = k + b$$

$$0 = 2k + b$$

$$b = -2k$$

$$1 = k - 2k \quad b = 2$$

$$k = -1 \quad b = 2$$

$$OO_1 (1; 1)$$

$$OO_2 (1+a; 1-a)$$

$$\begin{cases} 1+a=1 \\ 1-a=1 \end{cases}$$

$$a=0$$

$$\sqrt{(1+a-1)^2 + (1-a-1)^2} = \ell$$

$$\sqrt{a^2 + a^2} = \ell$$

$$\ell = \sqrt{2a^2} = |a|\sqrt{2}$$

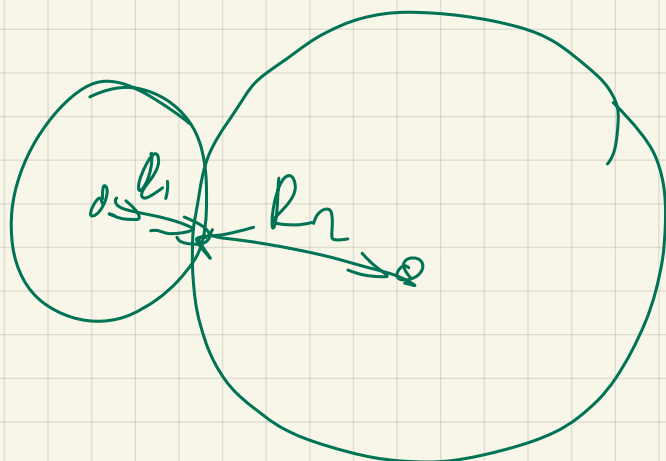
$$\ell = 2R = 2\sqrt{2}$$

$$|a|\sqrt{2} = 2\sqrt{2}$$

$$|a| = 2$$

$$a = \pm 2$$

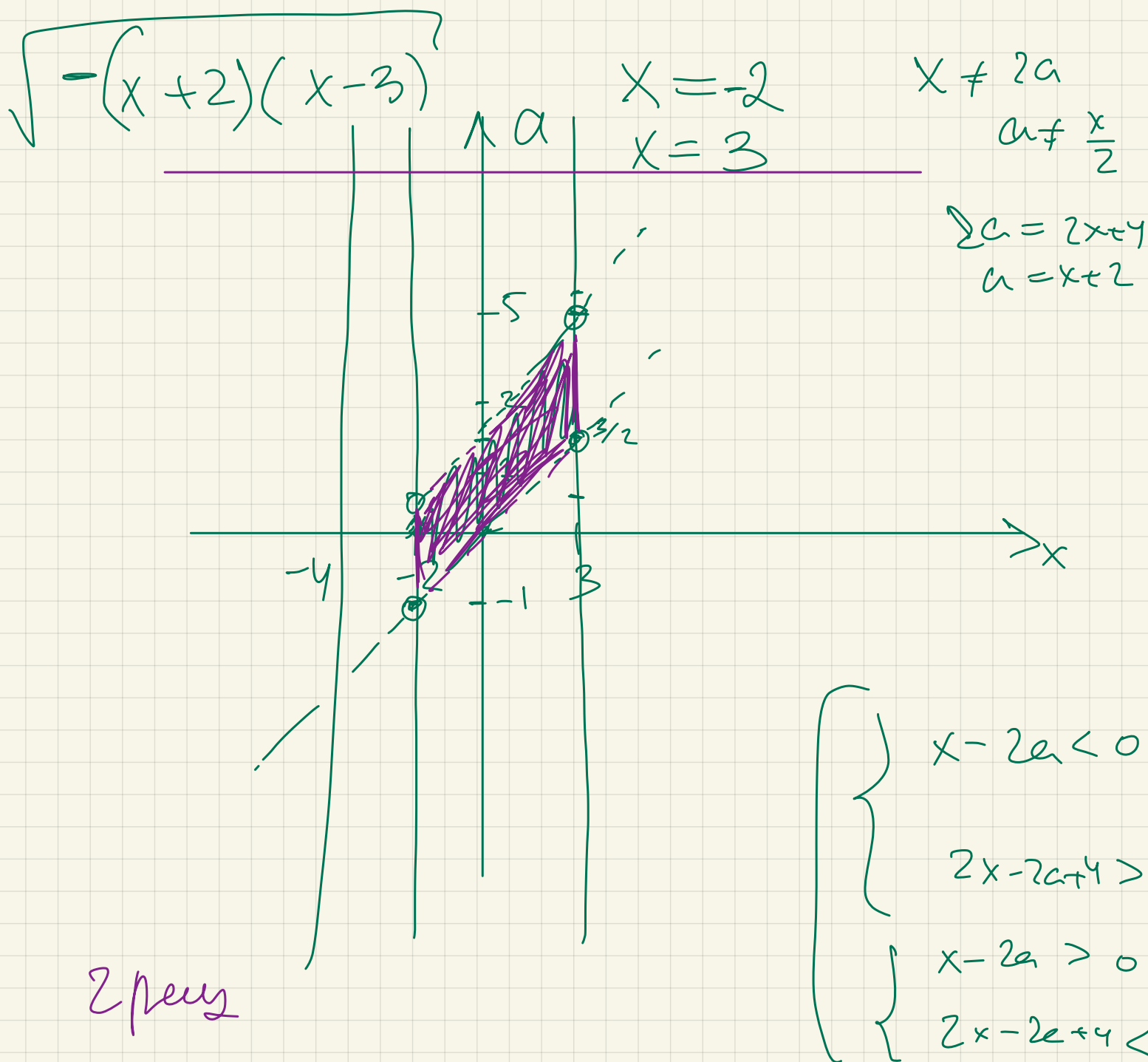
$$a \in (-2; 0) \cup (0; 2)$$



$$R_1 + R_2 = \ell$$

$$\frac{\sqrt{6+x-x^2}}{x-2a} \leq \frac{\sqrt{6+x-x^2}}{2x-2a+4} \quad x+4$$

$$\frac{\sqrt{6+x-x^2}(2x-2a+4-x+2a)}{(x-2a)(2x-2a+4)} \leq 0$$



$$a \in (-\infty; -1) \cup (5; +\infty)$$

$$x < 2c$$

$$2c > x+y$$

$$\begin{cases} x^2 - y^2 - 2(x+2y) - 3 = 0 \\ x^2 + y^2 - 2a(x+y) - 2x - 2a^2 + 6c \leq 0 \end{cases}$$

1 piece

$$(x-1)^2 + y^2$$

$$(x-1-a)^2 + (y-a)^2 \leq (2a-1)^2$$

$$-(1+a)$$

$$R = |2a-1|$$

$$y = -x-1$$

$$y = x-3$$

$$(1+a; a)$$

$$a=0$$

$a \uparrow$

$$y = kx + b$$

$$0 = k + b \rightarrow b = -k$$

$$1 = 2k + b$$

$$2k - k = 1$$

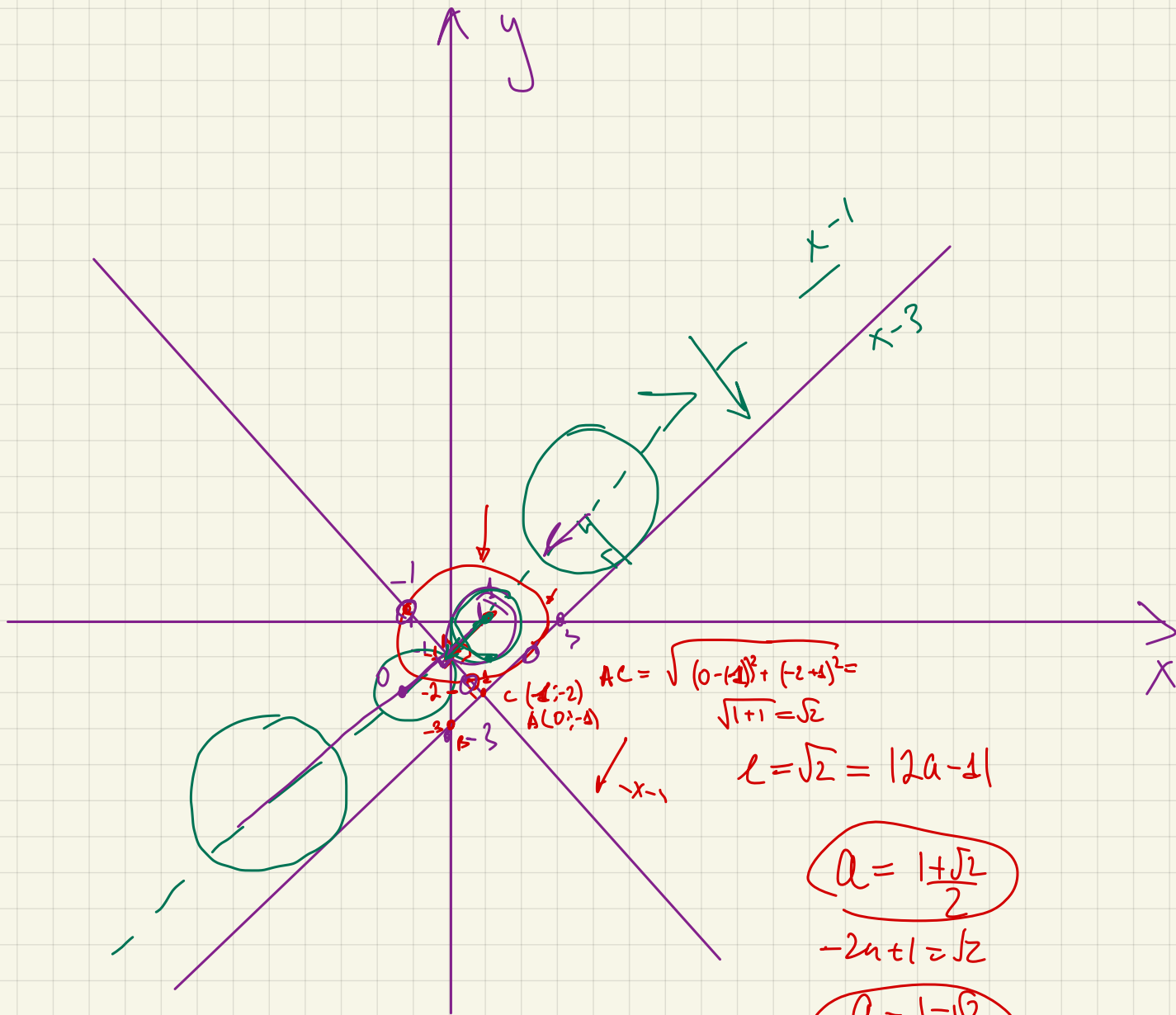
$$k = 1; b = -1$$

$$(1; 0)$$

$$a=1$$

$$(2; 1)$$

$$y = x-1$$



$$a = \frac{1+\sqrt{2}}{2}$$

$$-2a+1 = \sqrt{2}$$

$$a = \frac{1-\sqrt{2}}{2}$$

$$\begin{aligned}
 OA &\Rightarrow A(0, -1) \\
 (1+a, a) & \\
 \sqrt{(1+a)^2 + (a-1)^2} &= OA \\
 1+2a+a^2+a^2-2a+1 & \\
 \sqrt{2a^2+2} &= \sqrt{a^2+1} \sqrt{2}
 \end{aligned}$$

$$\sqrt{2} \sqrt{a^2+1} = |2a-1|$$

$$(a^2+1)2 = 4a^2-4a+1$$

$$R = |2a-1| < \sqrt{a^2+1} \sqrt{2}$$

$$\left| \frac{2(1+\sqrt{2})}{2} \right| < \sqrt{2} \left(\frac{1+2+2\sqrt{2}}{4} + 1 \right)$$

$$1+\sqrt{2} < \sqrt{2} \sqrt{\frac{7+2\sqrt{2}}{2}}$$

$$\begin{aligned}
 2+2\sqrt{2} &< \sqrt{2} \sqrt{7+2\sqrt{2}} \\
 \approx 4,8 &
 \end{aligned}$$

$$l \neq R$$

$$2a^2 - 4a - 1 = 0$$

$$\frac{4 \pm \sqrt{16 + 8}}{4} = \frac{4 \pm 2\sqrt{3}}{4} =$$

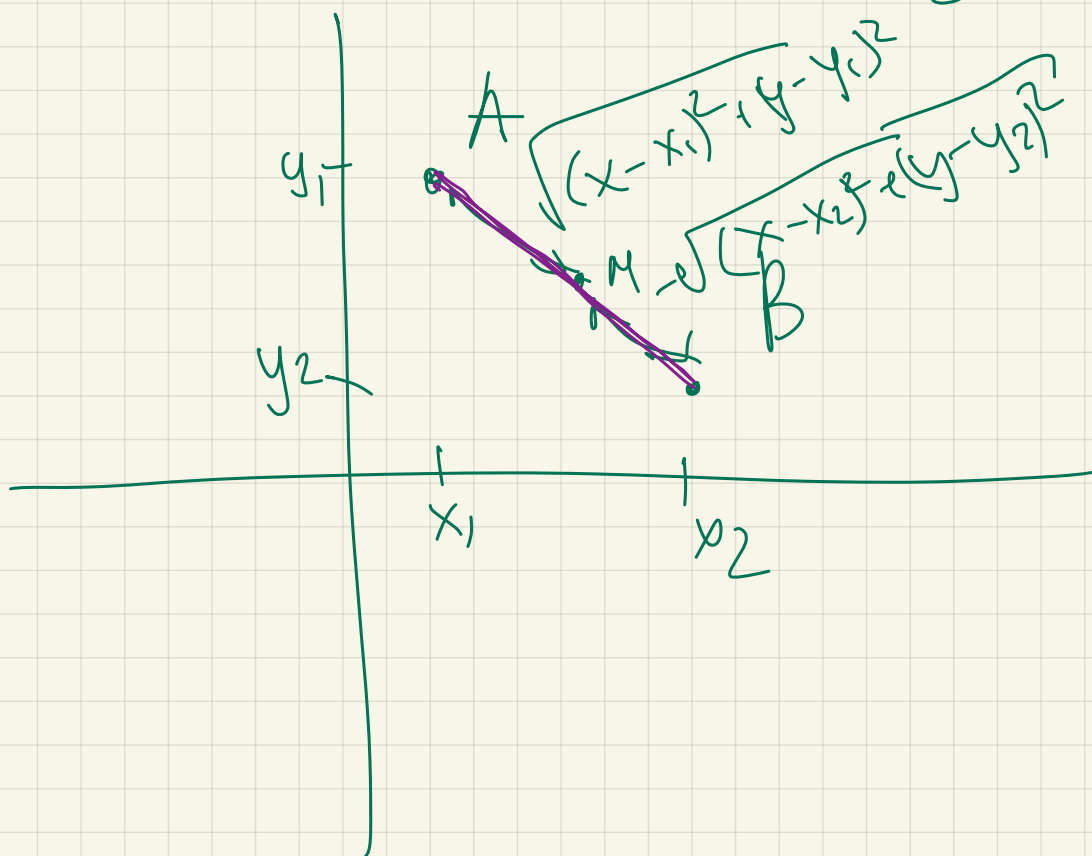
$$\frac{2 \pm \sqrt{3}}{2}$$

A $(x_1; y_1)$

B $(x_2; y_2)$

M $(x; y)$

$$\begin{aligned} & \sqrt{(x - x_1)^2 + (y - y_1)^2} + \\ & \sqrt{(x - x_2)^2 + (y - y_2)^2} = \\ & \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \end{aligned}$$



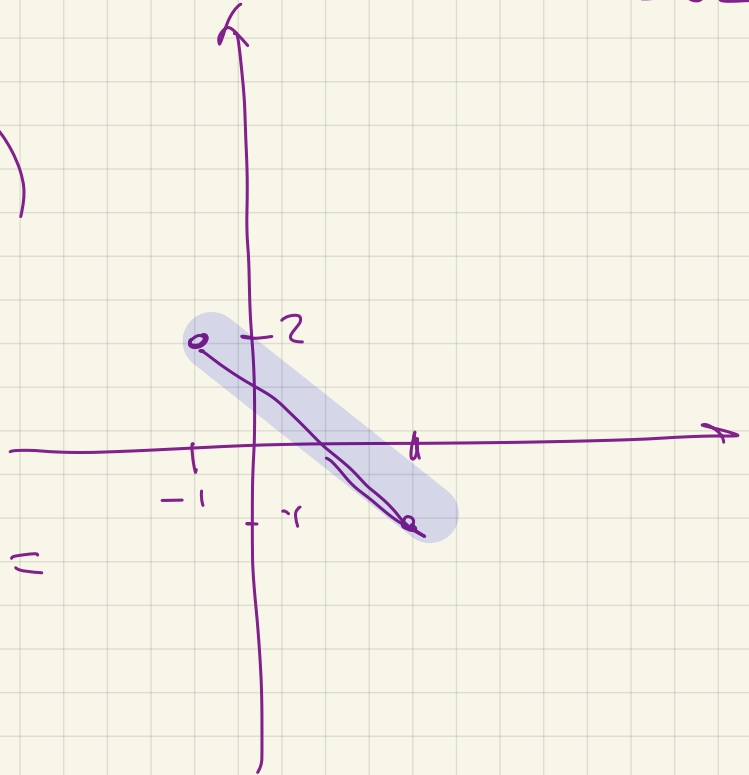
$$\sqrt{(x+1)^2 + (y-2)^2} + \sqrt{(x-2)^2 + (y+1)^2} = 3\sqrt{2}$$

$$A(-1; 2)$$

$$B(2; -1)$$

$$\sqrt{(2+1)^2 + 3^2} =$$

$$9+9 = \sqrt{18} = 3\sqrt{2}$$



$$\sqrt{(x-a)^2 + y^2} + \sqrt{x^2 + (y+a)^2} = |a\sqrt{2}|$$

$$(a; 0)$$

$$(0; -a)$$

$$\sqrt{a^2 + a^2} = |a|\sqrt{2}$$